

Calculus – Group Discussion

Week 5 – Definite Integrals and Differential Equations

ADSAI & CS / PR12

Learning goals (Week 5)

- Interpret the definite integral as accumulated change and signed area.
- Apply properties of definite integrals to simplify computations.
- Use the Fundamental Theorem of Calculus to evaluate integrals.
- Solve first-order separable differential equations.
- Solve first-order linear differential equations using integrating factors.

Riemann sums and the definite integral

A **Riemann sum** approximates the area under a curve by rectangles bounded in the range $[a, b]$. If $[a, b]$ is divided into n intervals $i = 1, 2, \dots, n$, then each interval has size

$$\Delta x = \frac{b - a}{n},$$

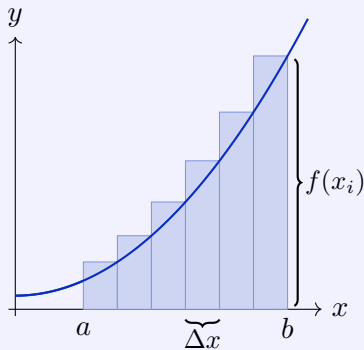
and the i -th interval is at $x_i = a + i \cdot \Delta x$.

The Riemann sum is

$$\sum_{i=1}^n f(x_i) \Delta x.$$

The **definite integral** is the limit of this sum:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x.$$



<https://www.desmos.com/calculator/tgyr42ezjq?lang=nl>

Fundamental Theorem of Calculus (FTC)

FTC Part 1. Let F be defined on I by

$$F(x) = \int_a^x f(t) dt.$$

Then F is differentiable on I , and $F'(x) = f(x)$ there:

$$\frac{d}{dx} \int_a^x f(t) dt = f(x).$$

FTC Part 2. If $F'(x) = f(x)$ on $[a, b]$, then

$$\int_a^b f(x) dx = F(b) - F(a).$$

This is the main tool for evaluating definite integrals.

Some properties of definite integrals

(a) Reversing limits of integration changes the sign of the integral

$$\int_a^b f(x)dx = - \int_b^a f(x)dx$$

(b) Linearity; if α and β are constants, then

$$\int_a^b (\alpha f(x) + \beta g(x))dx = \alpha \int_a^b f(x)dx + \beta \int_a^b g(x)dx$$

(c) An integral depends additively on the interval of integration

$$\int_a^b f(x)dx + \int_b^c f(x)dx = \int_a^c f(x)dx$$

(d) Chain rule extension of FTC

$$\frac{d}{dx} \int_a^{g(x)} f(t)dt = f(g(x))g'(x)$$

First-order differential equations

A **first-order differential equation** involves an unknown function $y(x)$ and its first derivative given by a known function $f(x, y)$.

$$\frac{dy}{dx} = f(x, y)$$

which describes the dynamics of processes and systems.

Separable equations. If $f(x, y) = g(x)h(y)$, we get

$$\frac{dy}{dx} = g(x)h(y).$$

and the variables can be separated in differential form:

$$\frac{1}{h(y)} dy = g(x) dx.$$

Integrating both sides and solving for y gives the solution.

First-order linear differential equations

A first-order **linear** differential equation is of the type

$$\frac{dy}{dx} + p(x)y = q(x) \quad (\text{linear in } y \text{ and } y')$$

where $p(x)$ and $q(x)$ are given functions, assumed to be continuous.

if $q(x) = 0 \implies$ **homogeneous**, separable and easy to solve to give

$$y = Ce^{-\mu(x)}, \text{ where } \mu(x) = \int p(x)dx$$

if $q(x) \neq 0 \implies$ **non-homogeneous**, solve using an *Integrating Factor*

Solving linear differential equations using integrating factors

To solve linear differential equations

$$\frac{dy}{dx} + p(x)y = q(x),$$

let $\mu(x) = \int p(x) dx$ and multiply the DE by $e^{\mu(x)}$.

$$e^{\mu(x)} \frac{dy}{dx} + e^{\mu(x)} p(x)y = e^{\mu(x)} q(x) \quad ((uv)' = uv' + vu')$$

$$\underbrace{\frac{d}{dx} [e^{\mu(x)} y]}_{\text{product rule}} = e^{\mu(x)} q(x) \quad (\text{Integrate both sides})$$

$$e^{\mu(x)} y = \int e^{\mu(x)} q(x) dx$$

$$y = e^{-\mu(x)} \int e^{\mu(x)} q(x) dx$$

Key takeaways

- A definite integral is the limit of Riemann sums and represents accumulated change / signed area.
- The Fundamental Theorem of Calculus connects area, antiderivatives, and differentiation.
- Remember the properties of definite integrals and how to apply them: limits reversal, linearity, interval additivity, chain rule extension of FTC.
- Separable differential equations can be solved by separating variables and integrating.
- Linear first-order equations have the form $y' + p(x)y = q(x)$.
 - if $q(x) = 0$, the DE is *homogeneous* (separable and easy to solve)
 - if $q(x) \neq 0$, the DE is *non-homogeneous* (solve using an integrating factor)